

Assignment of DSAIT4310

September 13, 2026

Consider the processed "SFHH" face-to-face contact network at a conference¹, which is given in the following format: each row "a b t" denotes an undirected contact (a temporal link) between node a and b at time step t. This contact network is sampled/measured once every 20 seconds during a conference. Thus, each time step has a duration of 20s, which is though not relevant for our analysis in this assignment. We denote this temporal network as G_{data} .

A. Explore properties of network G that is aggregated over all the $T = 3259$ steps. Specifically, the aggregated network G is composed of all the nodes that have ever appeared in the dataset and any two nodes are connected by a link if they have at least a contact over the whole period $[1, T]$.

Compute the following topological properties of G in 1)-6) without considering the weight of any link and the link weight property in 7).

- 1) What is the number of nodes N , the link density p and standard deviation of the degree $\sqrt{Var[D]}$?
- 2) Plot the degree distribution. Which network model, Erdős-Rényi (ER) random graphs or scale-free networks, could better model this network with respect to degree distribution? Explain how you arrived at your conclusion.
- 3) What is the degree correlation (assortativity) ρ_D ? What is its physical meaning?
- 4) What is the clustering coefficient C ?
- 5) What is the average hopcount $E[H]$ of the shortest paths between all node pairs? What is the diameter H_{max} ?
- 6) Has this network the small-world property? Justify your conclusion quantitatively (Hint: Lecture 2).
- 7) Consider further the weight of each link in G , which is the total number of contacts between the corresponding two nodes within $[1, T]$. Plot the probability density function (distribution) of the link weight (Choose the scales of the two axes and the bins/bin-size for the distribution such that the plot is insightful for interpretation). The probability density function $f_W(x)$ of the weight W of a link is defined as $f_W(x) = \lim_{\Delta x \rightarrow 0} \frac{Pr[x < W \leq x + \Delta x]}{\Delta x}$, the probability that the variable (or the percentage of links whose weight) is within each range or bin $(x, x + \Delta x]$ normalized by the size of the bin Δx . Does W follow a power-law distribution? Explain how you arrived at your conclusion.

Hint: All metrics computed for the network G are recommended to put into a table.

B. Information spreading on a temporal network

We consider the following information spreading process, which is actually a simplified Susceptible-Infected model but on a temporal network. Initially, at time $t = 0$, a single node s is infected meaning that this node possesses the information whereas all the other nodes are Susceptible, thus have not yet perceived the information. Node s is also called the seed of the information. Whenever an infected node i is in contact with a susceptible node j at any time step t , the susceptible node becomes infected during the same time step and could possibly infect other nodes only since the next time step via its contacts with susceptible nodes. Once a node becomes infected, it stays infected forever. For example, assume that the seed node s has its first contact

¹C. Cattuto et al., "Dynamics of person-to-person interactions from distributed rfid sensor networks", PloS one 5(7), e11596 (2010)

at time $t = 5$ and that contact is with node m . Although node s gets infected since $t = 0$, it infects a second node, i.e. node m only at $t = 5$ when it contacts m . Infection happens only when an infected node and a susceptible node are in contact. The number of infected nodes is non-decreasing over time.

Simulate the information spreading process on the given temporal network G_{data} for N iterations. Each iteration starts with a different seed node infected at $t = 0$ and ends at $t = T = 3259$ the last time step that the network is measured. Via the N iterations, we consider the spreading process that starts at every node $i \in [1, N]$. Record the total number of infected nodes $I(t)$ at each step t for each iteration.

8) Taking all the N iterations into count, plot the average number of infected nodes $E[I(t)]$ together with its error bar (standard deviation $\sqrt{Var[I(t)]}$) as a function of the time step t .

9) How influential a node is as a seed node could be quantified by the total number of nodes that have been infected by a specific time $t^{(l)} = 1200$ when that node is the sole seed node. This total number of infected nodes at $t^{(l)} = 1200$ is also called the influence of that node. Plot the influence of every node in descending order. Rank all nodes by influence and store the ranking in a vector $R = [R_{(1)}, R_{(2)}, \dots, R_{(N)}]$ where $R_{(i)}$ represents the index of the i -th most influential node and $R_{(1)}$ is the most influential node. (Note: you don't need to include this vector in your report.)

We will explore which nodal level network feature (centrality metric) in network G_{data} better predicts nodal influence described above. First, compute the total number of unique nodes that each node i has contacted during the time interval $[1, t^{(l)}]$. This is equivalent to the degree $d_i^{t^{(l)}}$ of node i in the aggregated network which is formed by aggregating the temporal network G_{data} over $[1, t^{(l)}]$. Based on this centrality metric derived for every node, construct the ordered vector $D = [D_{(1)}, D_{(2)}, \dots, D_{(N)}]$, where $D_{(i)}$ is the node index with the i -th highest degree in this aggregated network over $[1, t^{(l)}]$. How precisely a centrality metric (such as node degree in an aggregated network) predicts the ranking of nodes by influence could be quantified by the top f recognition rate $r_{RD}(f) = \frac{|R_f \cap D_f|}{|R_f|}$ where R_f and D_f are the sets of nodes ranking in the top f fraction according to influence and degree (in aggregated network) respectively and $|R_f| = fN$ is the number of nodes in R_f . (a) Plot $r_{RD}(f)$ as a function of f for $f = 0.05, 0.1, 0.15, \dots, 0.5$.

Note on ties: It is possible that multiple nodes may share the same rank, the selection of R_f or D_f may not be unique. In such case, $r_{RD}(f)$ must be computed as the average of 1000 iterations. In each iteration, R_f and D_f are sampled independently and uniformly at random from their corresponding possible choices and further used to compute the recognition rate. (This randomized tie-breaking procedure applies to all recognition rate calculations in this assignment.) Consider the degree sequence $\{d_1^{t^{(l)}} = 3, d_2^{t^{(l)}} = 1, d_3^{t^{(l)}} = 1, d_4^{t^{(l)}} = 1\}$ as an example. $D_{f=0.5}$ (top 2 nodes) has three possible choices $\{1, 2\}, \{1, 3\}$ and $\{1, 4\}$. A random choice $D_{f=0.5}$ from possible sets could also be obtained by firstly choosing the rank of nodes 2, 3 and 4 by degree as a randomized/reshuffled vector of $[2, 3, 4]$ since node 1 has rank 1 and afterwards selecting the 2 nodes out of the 4 with the highest rank.

Similarly, compute the degree $d_i^{t^s}$ of each node i in the aggregated network formed by aggregating G_{data} over $[1, t^s]$, where $t^s = 600$. Metric $d_i^{t^s}$ could also be used to estimate the ranking of nodes by influence. Construct the corresponding ordered vector $D' = [D'_{(1)}, D'_{(2)}, \dots, D'_{(N)}]$, where $D'_{(i)}$ is the node index with the i -th highest degree in the aggregated network over $[1, t^s]$. (b) Plot $r_{RD'}(f)$ as a function of f for $f = 0.05, 0.1, 0.15, \dots, 0.5$.

Consider a third nodal property: the time Z_i when node i makes its first contact in network D_{data} . Construct the ordered vector $Z = [Z_{(1)}, Z_{(2)}, \dots, Z_{(N)}]$, where $Z_{(i)}$ is the node index with the i -th shortest time to make its first contact and use Z to predict nodes' ranking by influence. (c) Plot the recognition rate $r_{RZ}(f)$ as a function of f for $f = 0.05, 0.1, 0.15, \dots, 0.5$. Plot the recognition rates obtained in (a), (b), and (c) on a single figure for direct comparison.

(d) Which centrality metric $d_i^{t^s}$, $d_i^{t^l}$ or Z_i best predict the ranking of nodes by influence and which performs the worst? Why?

10) The short-term influence of a node i is defined as the total number of nodes that have been infected at $t^{(s)} = 600$ when node i is the single seed node. Rank nodes by their short-term influence and record the ranking

in a vector $R' = [R'_{(1)}, R'_{(2)}, \dots, R'_{(N)}]$ where $R'_{(i)}$ is the index of the node with the i -th highest short-term influence. This short-term influence can also be used to estimate the influence (long-term) defined in 9). Plot the recognition rate $r_{RR'}(f)$ as a function of f in the same figure as in question (9). Can this short-term influence better predict (long-term) influence in ranking than those three centrality metrics studied in 9)? Why?

11) Another temporal network G_2 is given the attachment and is obtained as follows. G_2 is G_{data} where the time stamps describing when each temporal link (contact) appears in G_{data} are randomized. In other words, G_2 is constructed by copying all the temporal links $\{(a, b, t)\}$ from G_{data} but their time stamps t are randomly re-shuffled. The number of contacts between each node pair remains the same between G_{data} and G_2 . Now, perform the same analysis in 8)-10) for G_2 . In this case, the spreading process unfolds on G_2 and all centrality metrics and nodal influences are to be derived from G_2 .

a) Similar to question 8), plot the average number of infected nodes $E[I(t)]$ together with its error bar (standard deviation $\sqrt{Var[I(t)]}$) as a function of the time step t for network G_2 . Compare this result obtained from G_2 with your answer to 8) derived from G_{data} . What is the key difference? Please explain it.

b) Similar to 9) and 10), plot $r_{RD}(f)$, $r_{RD'}(f)$, $r_{RZ}(f)$ and $r_{RR'}(f)$ respectively for G_2 as a function of f . Compare your results here obtained from G_2 and answers to 9) and 10) obtained from G_{data} . What is the key difference? Explain the key difference.

Note: Questions asking 'why' or requesting explanations refer specifically to the causes behind your observations.